

READ DATASET

```
import pandas as pd
import numpy as np

df = pd.read_csv('wan_dataset.csv')
```

DATA ANALYSIS

```
df.info()
print(df.shape)
print(df.columns)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   timestamp             2000 non-null  int64
 1   link_id               2000 non-null  object
 2   source_device         2000 non-null  object
 3   destination_device    2000 non-null  object
 4   source_interface      2000 non-null  object
 5   destination_interface  2000 non-null  object
 6   link_utilization      2000 non-null  int64
 7   latency_ms            2000 non-null  int64
 8   packet_loss           2000 non-null  float64
 9   throughput_mbps       2000 non-null  int64
10   qos_priority          2000 non-null  int64
11   reachability          2000 non-null  int64
12   traffic_type          2000 non-null  object
13   anomaly_flag          2000 non-null  int64
dtypes: float64(1), int64(7), object(6)
memory usage: 218.9+ KB
(2000, 14)
Index(['timestamp', 'link_id', 'source_device', 'destination_device',
      'source_interface', 'destination_interface', 'link_utilization',
      'latency_ms', 'packet_loss', 'throughput_mbps', 'qos_priority',
      'reachability', 'traffic_type', 'anomaly_flag'],
      dtype='object')
```

DESCRIPTION

```
df.describe()
```

|       | timestamp   | link_utilization | latency_ms  | packet_loss | throughput_mbps | qos_priority | reachability | anomaly_flag |
|-------|-------------|------------------|-------------|-------------|-----------------|--------------|--------------|--------------|
| count | 2000.000000 | 2000.000000      | 2000.000000 | 2000.000000 | 2000.000000     | 2000.000000  | 2000.000000  | 2000.000000  |
| mean  | 99.500000   | 49.905000        | 18.941500   | 0.981000    | 204.797000      | 3.000000     | 0.960000     | 0.020000     |
| std   | 57.748744   | 28.220067        | 10.552334   | 0.577413    | 211.489704      | 1.414567     | 0.196008     | 0.140035     |
| min   | 0.000000    | 10.000000        | 5.000000    | 0.000000    | -99.000000      | 1.000000     | 0.000000     | 0.000000     |
| 25%   | 49.750000   | 22.000000        | 8.000000    | 0.480000    | -7.000000       | 2.000000     | 1.000000     | 0.000000     |
| 50%   | 99.500000   | 50.000000        | 18.000000   | 1.000000    | 210.000000      | 3.000000     | 1.000000     | 0.000000     |
| 75%   | 149.250000  | 78.000000        | 29.250000   | 1.480000    | 417.000000      | 4.000000     | 1.000000     | 0.000000     |
| max   | 199.000000  | 89.000000        | 35.000000   | 2.000000    | 499.000000      | 5.000000     | 1.000000     | 1.000000     |

COUNT OF NULL VALUES

```
n = df.isnull()
n.count()
```

|                       | 0    |
|-----------------------|------|
| timestamp             | 2000 |
| link_id               | 2000 |
| source_device         | 2000 |
| destination_device    | 2000 |
| source_interface      | 2000 |
| destination_interface | 2000 |
| link_utilization      | 2000 |
| latency_ms            | 2000 |
| packet_loss           | 2000 |
| throughput_mbps       | 2000 |
| qos_priority          | 2000 |
| reachability          | 2000 |
| traffic_type          | 2000 |
| anomaly_flag          | 2000 |

dtype: int64

MISSING VALUE HANDLING

df.fillna(0)

|      | timestamp | link_id | source_device | destination_device | source_interface | destination_interface | link_utilization | latency_ms |
|------|-----------|---------|---------------|--------------------|------------------|-----------------------|------------------|------------|
| 0    | 0         | link_1  | host_1        | switch_1           | interface_1      | interface_1           | 50               | 10         |
| 1    | 0         | link_2  | host_2        | switch_3           | interface_2      | interface_4           | 83               | 20         |
| 2    | 0         | link_3  | host_3        | switch_1           | interface_3      | interface_7           | 86               | 30         |
| 3    | 0         | link_4  | host_4        | switch_3           | interface_4      | interface_10          | 55               | 40         |
| 4    | 0         | link_5  | host_5        | switch_1           | interface_5      | interface_1           | 19               | 50         |
| ...  | ...       | ...     | ...           | ...                | ...              | ...                   | ...              | ...        |
| 1995 | 199       | link_6  | host_1        | switch_2           | interface_8      | interface_11          | 40               | 60         |
| 1996 | 199       | link_7  | host_2        | switch_4           | interface_9      | interface_2           | 77               | 70         |
| 1997 | 199       | link_8  | host_3        | switch_2           | interface_10     | interface_5           | 89               | 80         |
| 1998 | 199       | link_9  | host_4        | switch_4           | interface_11     | interface_8           | 64               | 90         |
| 1999 | 199       | link_10 | host_5        | switch_2           | interface_12     | interface_11          | 26               | 100        |

2000 rows × 14 columns

DATA SORTING

```
cf = df[df['packet_loss'] < 0.50]
cf.head(8)
```

|  | timestamp | link_id | source_device | destination_device | source_interface | destination_interface | link_utilization | latency_ms |
|--|-----------|---------|---------------|--------------------|------------------|-----------------------|------------------|------------|
|  | 0         | link_1  | host_1        | switch_1           | interface_1      | interface_1           | 50               | 35         |
|  | 2         | link_3  | host_3        | switch_1           | interface_3      | interface_7           | 86               | 13         |
|  | 5         | link_6  | host_6        | switch_3           | interface_6      | interface_4           | 11               | 24         |
|  | 7         | link_8  | host_2        | switch_3           | interface_8      | interface_10          | 76               | 31         |
|  | 10        | link_1  | host_2        | switch_2           | interface_3      | interface_2           | 53               | 34         |
|  | 15        | link_6  | host_1        | switch_4           | interface_8      | interface_5           | 12               | 24         |
|  | 23        | link_4  | host_6        | switch_1           | interface_8      | interface_12          | 47               | 5          |
|  | 28        | link_9  | host_5        | switch_3           | interface_1      | interface_3           | 87               | 16         |

FREQUENCY LISTING

df['reachability'].value\_counts()

|              | count |
|--------------|-------|
| reachability |       |
| 1            | 1920  |
| 0            | 80    |

dtype: int64

df['qos\_priority'].value\_counts()

|              | count |
|--------------|-------|
| qos_priority |       |
| 1            | 400   |
| 2            | 400   |
| 3            | 400   |
| 4            | 400   |
| 5            | 400   |

dtype: int64

DATA SORTING

```
sorted_rows = df.sort_values(by="latency_ms", ascending=True)
print(sorted_rows.head(5))

sorted_columns = df[sorted(df.columns)]
print(sorted_columns.head(5))

print(df.iloc[0:5, 0:4])

print(df.loc[0:4, ["link_id", "latency_ms", "packet_loss"]])
```

|      | timestamp             | link_id          | source_device | destination_device | source_interface | \ |
|------|-----------------------|------------------|---------------|--------------------|------------------|---|
| 13   | 1                     | link_4           | host_5        | switch_4           | interface_6      |   |
| 3    | 0                     | link_4           | host_4        | switch_3           | interface_4      |   |
| 1986 | 198                   | link_7           | host_1        | switch_3           | interface_7      |   |
| 23   | 2                     | link_4           | host_6        | switch_1           | interface_8      |   |
| 1976 | 197                   | link_7           | host_6        | switch_2           | interface_5      |   |
|      | destination_interface | link_utilization | latency_ms    | packet_loss        | \                |   |
| 13   | interface_11          | 51               | 5             | 1.88               |                  |   |
| 3    | interface_10          | 55               | 5             | 1.60               |                  |   |
| 1986 | interface_1           | 74               | 5             | 0.64               |                  |   |
| 23   | interface_12          | 47               | 5             | 0.16               |                  |   |
| 1976 | interface_12          | 71               | 5             | 0.36               |                  |   |
|      | throughput_mbps       | qos_priority     | reachability  | traffic_type       | anomaly_flag     |   |
| 13   | 221                   | 5                | 1             | http               | 0                |   |

```
3          242          4          1  file_transfer          0
1986        454          5          1          http          0
23          200          1          1          video          0
1976        442          4          1  file_transfer          0
  anomaly_flag destination_device destination_interface latency_ms link_id \
0          1          switch_1          interface_1          35 link_1
1          0          switch_3          interface_4          28 link_2
2          0          switch_1          interface_7          13 link_3
3          0          switch_3          interface_10         5 link_4
4          0          switch_1          interface_1          10 link_5

  link_utilization packet_loss qos_priority reachability source_device \
0          50          0.0          1          0          host_1
1          83          1.2          2          1          host_2
2          86          0.4          3          1          host_3
3          55          1.6          4          1          host_4
4          19          0.8          5          1          host_5

  source_interface throughput_mbps timestamp traffic_type
0  interface_1          200          0          http
1  interface_2          452          0          video
2  interface_3          472          0          voip
3  interface_4          242          0  file_transfer
4  interface_5          -27          0          http

  timestamp link_id source_device destination_device
0          0 link_1          host_1          switch_1
1          0 link_2          host_2          switch_3
2          0 link_3          host_3          switch_1
3          0 link_4          host_4          switch_3
4          0 link_5          host_5          switch_1

  link_id latency_ms packet_loss
0 link_1          35          0.0
1 link_2          28          1.2
2 link_3          13          0.4
3 link_4           5          1.6
4 link_5          10          0.8
```

CONDITION ROW DISPLAY

```
df.loc[10, ['reachability', 'throughput_mbps']]
```

```

      10
reachability    1
throughput_mbps 220
```

dtype: object

```
df.loc[200, ['qos_priority']]
```

```

      200
qos_priority    1
```

dtype: object

```
df.loc[1000, ['link_id', 'link_utilization', 'traffic_type']]
```

```

      1000
link_id    link_1
link_utilization    28
traffic_type    http
```

dtype: object

MIN AND MAX VALUES

```
df.max()
```

|                       | 0           |
|-----------------------|-------------|
| timestamp             | 199         |
| link_id               | link_9      |
| source_device         | host_6      |
| destination_device    | switch_4    |
| source_interface      | interface_9 |
| destination_interface | interface_9 |
| link_utilization      | 89          |
| latency_ms            | 35          |
| packet_loss           | 2.0         |
| throughput_mbps       | 499         |
| qos_priority          | 5           |
| reachability          | 1           |
| traffic_type          | voip        |
| anomaly_flag          | 1           |

dtype: object

df.min()

|                       | 0             |
|-----------------------|---------------|
| timestamp             | 0             |
| link_id               | link_1        |
| source_device         | host_1        |
| destination_device    | switch_1      |
| source_interface      | interface_1   |
| destination_interface | interface_1   |
| link_utilization      | 10            |
| latency_ms            | 5             |
| packet_loss           | 0.0           |
| throughput_mbps       | -99           |
| qos_priority          | 1             |
| reachability          | 0             |
| traffic_type          | file_transfer |
| anomaly_flag          | 0             |

dtype: object

GROUP BY

df.groupby('traffic\_type')['packet\_loss'].count()

|               | packet_loss |
|---------------|-------------|
| traffic_type  |             |
| file_transfer | 500         |
| http          | 500         |
| video         | 500         |
| voip          | 500         |

dtype: int64

```
df.groupby(['reachability', 'throughput_mbps'])['latency_ms'].count()
```

|              |                 | latency_ms |
|--------------|-----------------|------------|
| reachability | throughput_mbps |            |
| 0            | -99             | 1          |
|              | -96             | 1          |
|              | -95             | 2          |
|              | -89             | 1          |
|              | -88             | 2          |
| ...          | ...             | ...        |
| 1            | 495             | 11         |
|              | 496             | 15         |
|              | 497             | 15         |
|              | 498             | 21         |
|              | 499             | 48         |

663 rows × 1 columns

dtype: int64

AGGREGATION

```
df.groupby('throughput_mbps').agg({'packet_loss': 'mean'})
```

|                 |     | packet_loss |
|-----------------|-----|-------------|
| throughput_mbps |     |             |
| -99             |     | 0.960000    |
| -98             |     | 1.030476    |
| -97             |     | 0.770667    |
| -96             |     | 1.128571    |
| -95             |     | 1.006667    |
| ...             | ... | ...         |
| 495             |     | 0.943333    |
| 496             |     | 1.154667    |
| 497             |     | 0.880000    |
| 498             |     | 0.996522    |
| 499             |     | 1.052000    |

595 rows × 1 columns

```
df.groupby('qos_priority').agg({'latency_ms': 'count'})
```

|              |  | latency_ms |
|--------------|--|------------|
| qos_priority |  |            |
| 1            |  | 400        |
| 2            |  | 400        |
| 3            |  | 400        |
| 4            |  | 400        |
| 5            |  | 400        |

CONDITIONAL SELECTION

```
df['anomaly_flag'].head(10)
```

|   | anomaly_flag |
|---|--------------|
| 0 | 1            |
| 1 | 0            |
| 2 | 0            |
| 3 | 0            |
| 4 | 0            |
| 5 | 0            |
| 6 | 0            |
| 7 | 0            |
| 8 | 0            |
| 9 | 0            |

**dtype:** int64

```
a = df[df['packet_loss'] == 0.4]
a['packet_loss'].sum()
```

```
np.float64(16.0)
```

Correlation between any two columns

```
correlation = df["link_utilization"].corr(df["latency_ms"])

print("Correlation:", correlation)
```

```
Correlation: 0.189331720160936
```

Transformation on dataset

```
df["latency_normalized"] = (
    (df["latency_ms"] - df["latency_ms"].min()) /
    (df["latency_ms"].max() - df["latency_ms"].min())
)

print(df[["latency_ms", "latency_normalized"]].head())
```

|   | latency_ms | latency_normalized |
|---|------------|--------------------|
| 0 | 35         | 1.000000           |
| 1 | 28         | 0.766667           |
| 2 | 13         | 0.266667           |
| 3 | 5          | 0.000000           |
| 4 | 10         | 0.166667           |

Joining, Merging and Concatenation

```
df1 = df[["link_id", "source_device", "latency_ms"]].drop_duplicates("link_id")
df2 = df[["link_id", "throughput_mbps", "packet_loss"]].drop_duplicates("link_id")

joined = df1.set_index("link_id").join(
    df2.set_index("link_id"),
    how="inner"
)

print("Joined DataFrame:")
print(joined.head())

merged = pd.merge(
    df1,
    df2,
    on="link_id",
    how="inner"
)

print("Merged DataFrame:")
```

```
print(merged.head())

concat_rows = pd.concat(
    [df.head(5), df.tail(5)],
    ignore_index=True
)

print("Concatenated DataFrame:")
print(concat_rows)
```

|        |        |    |     |     |
|--------|--------|----|-----|-----|
| link_3 | host_3 | 13 | 472 | 0.4 |
| link_4 | host_4 | 5  | 242 | 1.6 |
| link_5 | host_5 | 10 | -27 | 0.8 |

Merged DataFrame:

|   | link_id | source_device | latency_ms | throughput_mbps | packet_loss |
|---|---------|---------------|------------|-----------------|-------------|
| 0 | link_1  | host_1        | 35         | 200             | 0.0         |
| 1 | link_2  | host_2        | 28         | 452             | 1.2         |
| 2 | link_3  | host_3        | 13         | 472             | 0.4         |
| 3 | link_4  | host_4        | 5          | 242             | 1.6         |
| 4 | link_5  | host_5        | 10         | -27             | 0.8         |

Concatenated DataFrame:

|   | timestamp | link_id | source_device | destination_device | source_interface | \ |
|---|-----------|---------|---------------|--------------------|------------------|---|
| 0 | 0         | link_1  | host_1        | switch_1           | interface_1      |   |
| 1 | 0         | link_2  | host_2        | switch_3           | interface_2      |   |
| 2 | 0         | link_3  | host_3        | switch_1           | interface_3      |   |
| 3 | 0         | link_4  | host_4        | switch_3           | interface_4      |   |
| 4 | 0         | link_5  | host_5        | switch_1           | interface_5      |   |
| 5 | 199       | link_6  | host_1        | switch_2           | interface_8      |   |
| 6 | 199       | link_7  | host_2        | switch_4           | interface_9      |   |
| 7 | 199       | link_8  | host_3        | switch_2           | interface_10     |   |
| 8 | 199       | link_9  | host_4        | switch_4           | interface_11     |   |
| 9 | 199       | link_10 | host_5        | switch_2           | interface_12     |   |

|   | destination_interface | link_utilization | latency_ms | packet_loss | \ |
|---|-----------------------|------------------|------------|-------------|---|
| 0 | interface_1           | 50               | 35         | 0.00        |   |
| 1 | interface_4           | 83               | 28         | 1.20        |   |
| 2 | interface_7           | 86               | 13         | 0.40        |   |
| 3 | interface_10          | 55               | 5          | 1.60        |   |
| 4 | interface_1           | 19               | 10         | 0.80        |   |
| 5 | interface_11          | 40               | 9          | 1.72        |   |
| 6 | interface_2           | 77               | 5          | 0.92        |   |
| 7 | interface_5           | 89               | 15         | 0.12        |   |
| 8 | interface_8           | 64               | 29         | 1.32        |   |
| 9 | interface_11          | 26               | 34         | 0.52        |   |

|   | throughput_mbps | qos_priority | reachability | traffic_type  | anomaly_flag | \ |
|---|-----------------|--------------|--------------|---------------|--------------|---|
| 0 | 200             | 1            | 0            | http          | 1            |   |
| 1 | 452             | 2            | 1            | video         | 0            |   |
| 2 | 472             | 3            | 1            | voip          | 0            |   |
| 3 | 242             | 4            | 1            | file_transfer | 0            |   |
| 4 | -27             | 5            | 1            | http          | 0            |   |
| 5 | 224             | 5            | 1            | http          | 0            |   |
| 6 | 464             | 1            | 1            | video         | 0            |   |
| 7 | 461             | 2            | 1            | voip          | 0            |   |
| 8 | 218             | 3            | 1            | file_transfer | 0            |   |
| 9 | -42             | 4            | 1            | http          | 0            |   |

|   | new_column                 | latency_normalized |
|---|----------------------------|--------------------|
| 0 | interface_1 - interface_1  | 1.000000           |
| 1 | interface_2 - interface_4  | 0.766667           |
| 2 | interface_3 - interface_7  | 0.266667           |
| 3 | interface_4 - interface_10 | 0.000000           |
| 4 | interface_5 - interface_11 | 0.444444           |